# PROPOSAL TO SIMPLIFY SERIES 7000 COMPUTER - INSTRUCTION CODES AND FORMATS.

CHANGES TO BE MADE TO THE MACHINE DEFINITION AS SUPPLIED BY ENGINEERING PLANNING ARE LISTED IN SUMMARY FORM.

TO SIMPLIFY THE INSTRUCTION FORMAT SOME INSTRUCTION DELETION, OR OPERATIONAL FACILITY DELETION SEEMED NECESSARY, DESIREABLE OR BOTH.

INSTRUCTION DELETION:

- RE NAME
- STORE ADDRESS
- LOAD INDEX
- LOAD REFILL
- STORE INDEX
- STORE REFILL
- ~~FULL WORD NO-OP~~
- ~~INTERRUPTIVE OPN~~
- STORE ZERO

OPERATIONAL FACILITY DELETION:

- DIMINISH VARIATION OF ADVANCE MODIFIER OF COUNT & BR. INSTRUCTIONS XMIT ½ WORD IMMEDIATE COUNT REDUCED FROM MAX of 32 TO MAX OF 16. PROGRESSIVE INDEXING IN ITS ENTIRETY.

- CHANGE 1 BRANCH INDICATOR FROM ½ WORD TO FULL WORD INSTRUCTION.

OTHER POTENTIAL - POSSIBLE CHANGES - CANDIDATES FOR ELIMINATION.

ELIMINATE:

- F.P. MPY
- F.P. DIVIDE
- F.P. ADD MAGN.
- F.P. ADD
- CONT COMP.

ELIMINATE:

- USE OF SIGNS FOR INDEXING ARITHMETIC
- ? USE OF ADDER FOR GENERATING INTERRUPT ADDRESS
- ? NECESSITY OF FULL COMPARE ON ADDRESS MATCH MAKE IT AN OPTIONAL FEATURE

# PROPOSAL TO SIMPLIFY SERIES 7000 COMPUTER

THE ADVANTAGES OF ADOPTING THE RECOMMENDED SIMPLIFICATIONS.

1. THE NUMBER OF DISTINCT INDIVIDUAL FORMATS IS REDUCED FROM 12 to 8
   a) THUS BUYING SIMPLIFICATION FOR PROGRAMMER ETC.
   b) THE GATING ON THE Z REGISTER IS CONSIDERABLY REDUCED, RESULTING IN SUBSTANTIAL TRANSISTOR SAVINGS.
   c) THE SIMILICITY AFFORDED BY THE NEW FORMAT FIELD MAKES PARITY CHECKING IN THE I BOX ATTRACTIVE, RESULTING IN SUBSTANTIAL TRANSISTOR SAVINGS IN THE CHECKING AREA.
   d) A FEW LEVELS OF LOGIC CAN BE ELIMINATED BUT IT IS NOT YET EVIDENT THAT THIS CAN RESULT IN A FASTER I BOX CYCLE

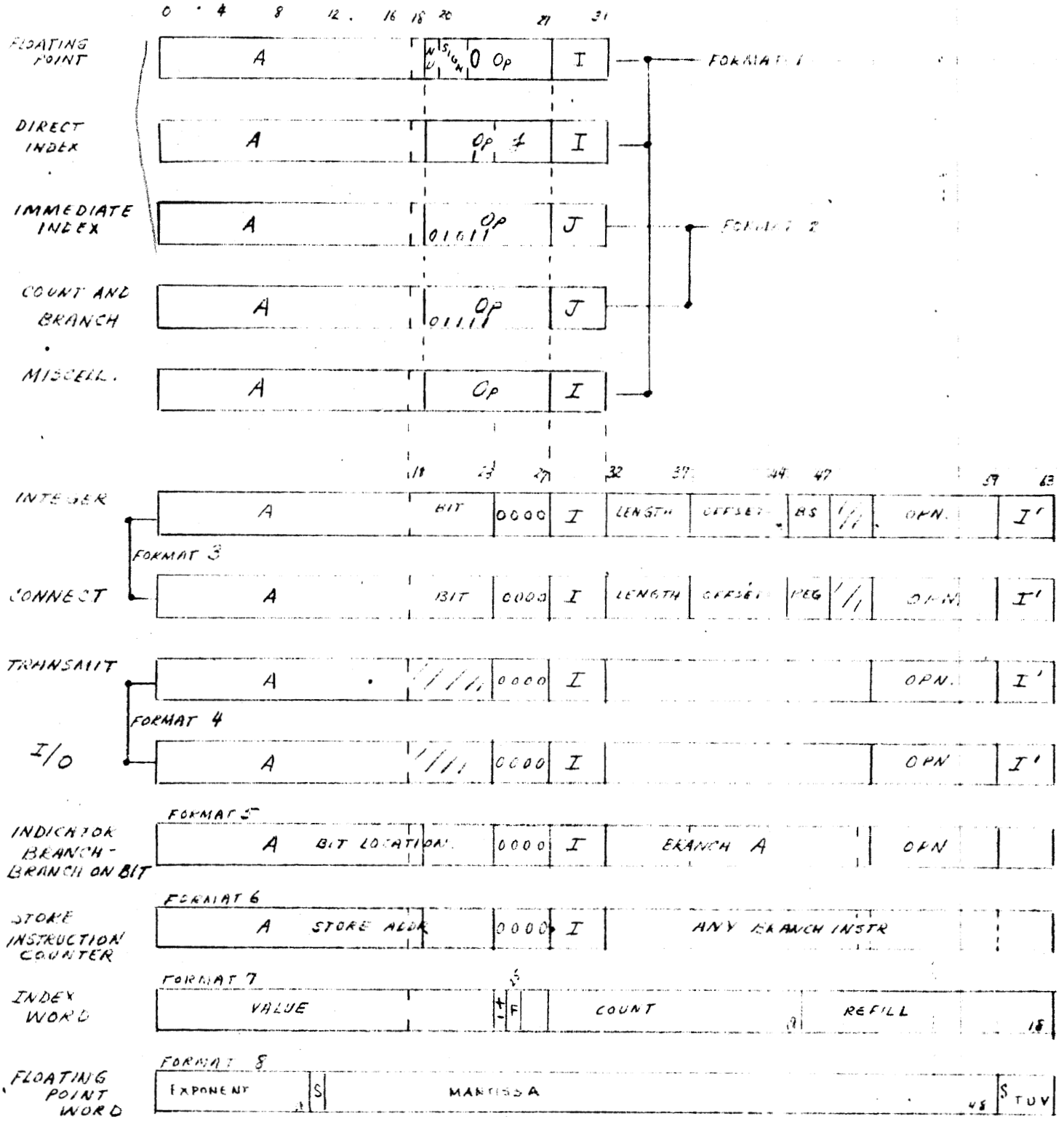
2. THE POWER OR POTENCY OF THE COMPUTER, VOCABULARY OR CAPABILITIES IS VIRTUALLY UNAFFECTED.

NOTE: SOME OF THE INSTRUCTIONS ELIMINATED WERE SELECTED ON A HASTY JUDGEMENT SO THE POSSIBILITY EXISTS THAT A SUBSTITUTION OF A RETAINED INSTRUCTION FOR ONE LISTED AS ELIMINATED MIGHT PROVIDE A MORE ACCEPTABLE COMPUTER SIMPLIFICATION PROPOSAL.

W Wheelwright

PROPOSAL 0411

4-11-58




THIS PROPOSAL OFFERS 8 FORMATS
PRESENT SYSTEM HAS 12 FORMATS

FLOATING POINT INSTRUCTIONS:

| { NU / SIGN } xxxxxx | | xxxxxx | |
|---|---|---|---|
| 000001 | LOAD | 010000 | (NOT TO BE USED) |
| 000010 | AUGMNT | 010001 | LOAD WI FLAG |
| 000011 | ADD MANT. | 010010 | INTRCH AUGMT |
| 000100 | ADD MEM. | 010011 | SHIFT MANT. |
| 000101 | STORE | 010100 | ADD TO MEM MAGN. |
| * 000110 | MPY | 010101 | ROUND & STR. |
| * 000111 | DIVIDE | * 010110 | ADD MAGN. |
| 001000 | ADD DBL. | 010111 | INTERCH DIV. |
| 001001 | LOAD DBL. | 011000 | ADD DBL MAGN. |
| 001010 | COMPARE | * 011001 | ADD |
| 001011 | ADD EXPONENT. | 011010 | —— SPARE |
| 001100 | CUM. MPY | 011011 | ADD EXP IMMED. |
| 001101 | STR. LW. ORDR | 011100 | LOAD CUM. |
| 001110 | MPY. DBL. | 011101 | BORROW & STR. |
| 001111 | DIV. WI. REM. | 011110 | SQ. RT. & STR |
| 000000 | (NOT TO BE USED) | 011111 | —— SPARE |

* CANDIDATES FOR ELIMINATION - WOULD PROVIDE 4 EXTRA SPARE OPN CODES

DIRECT INDEX INSTRUCTIONS:

| xxxxxJJJJ | |
|---|---|
| 000100000 | (NOT TO BE USED) |
| 00010JJJJ | LOAD VALUE |
| 00110 | LOAD COUNT |
| 01010 | COMP VALUE |
| 01110 | INCREMENT |
| 10010 | INCR. & CNT |
| 10110 | INCR, CNT & REFILL |
| 11010 | STORE VALUE |
| 11110 | STORE COUNT |
| 00011 | COMPARE COUNT |
| 00111 | LOAD INDIRECT |

TWO INSTRUCTIONS FALL IN SPECIAL CATEGORY

NOTE: J's VARY FROM 1 TO 15 SPECIFYING INDEX REGISTER TO BE OPERATED UPON.

ELIMINATED FROM THIS GROUP:

RENAME
—— STORE ADDRESS

- LOAD INDEX
- LOAD REFILL
- STORE INDEX
- STORE REFILL

IMMEDIATE INDEX INSTRUCTIONS:

| | |
|---|---|
| xxxxxxxxx | |
| 01011 0000 | (NOT TO BE USED) |
| 01011 0001 | LOAD VALUE |
| 0010 | LOAD COUNT |
| 0011 | LOAD REFILL |
| 0100 | COMPARE VALUE |
| 0101 | INCREMENT |
| 0110 | INCR. & CNT |
| 0111 | INCR.-CNT & REFILL |
| 1000 | SUBTR FR. COUNT |
| 1001 | LOAD VAL NEG. |
| 1010 | COMPARE COUNT |
| 1011 | ADD TO COUNT. |
| 1100 | COMP. VAL. NEG. |
| 1101 | DECREMENT |
| 1110 | DECR. & COUNT |
| 1111 | DECR.-CNT & REFILL |

ELIMINATED FROM THIS GROUP BUT NOT FROM THE COMPUTER: GEOMETRIC LOAD

COUNT & BRANCH INSTRUCTIONS:

| | | |
|---|---|---|
| xxxxxxxxx | | |
| 01111 0000 | (NOT TO BE USED) | COUNT & BRANCH UN INDEXED |
| 01111 xx01 | NO ADVANCE | |
| xx10 | HALF ADVANCE | |
| xx11 | FULL ADVANCE | |
| x0xx | COUNT & BRANCH | |
| x1xx | COUNT, REFILL & BRANCH | |
| 0xxx | BRANCH IF OFF | |
| 1xxx | BRANCH IF ON | |
| 10011 xxxx | COUNT & BRANCH FORM INDEXED BY X1 | |

ELIMINATED FROM THIS GROUP IS THE DIMINISH VARIATION OF THE ADVANCE MODIFIER.

MISCELLANEOUS & UNCONDITIONAL BRANCH INSTRUCTIONS:

| | |
|---|---|
| xxxxxxxxx | |
| 101110000 | (NOT TO BE USED) |
| 101110001 | BRANCH |
| 0010 | BRANCH ENABLE |
| 0011 | BRANCH DISABLE |
| 0100 | NO-OP |
| 0101 | SET & WAIT |
| 0110 | RELATIVE BRANCH |
| 0111 | —— SPARE |
| 1000 | REFILL |
| 1001 | COND REFILL |
| 1010 | INDIRECT EXECUTE |
| 1011 | EXECUTE |
| 1100 | GEOMETRIC LOAD |
| 1101 | —— SPARE |
| 1110 | —— SPARE |
| 1111 | —— SPARE |

ELIMINATED FROM THIS GROUP:
NO-OP FULL WD
INTERPRETIVE OPN
STORE ZERO

ADDED TO THIS GROUP: GEOMETRIC LOAD.

· ELIMINATED FROM HALF WORD INSTRUCTION SET -- INDICATOR BRANCH ←

SPARE HALF WORD INSTRUCTION OPN. CODES:

| | |
|---|---|
| xxxxxxxxx | |
| 11011xxxx | — 15 CODES (0000 - NOT TO BE USED) |
| 11111xxxx | — 15 CODES |

INTEGER 24 25 26 27

INSTRUCTIONS : 0 0 0 0 ~51 . . . . . 59

0 0 0 0 x x x x x x x x x

x x x x 0000 (NOT TO BE USED)

0 00001 AUGMENT

00010 LOAD

00011 ADD

00100 ADD TO MEM

00101 AUG TO MEM

00110 STORE

00111 ADD TO MAGN

01000 COMPARE

01001 COUNT TO MEM

* 01010 CONT COMP

01011 ——— SPARE

01100 SUM MPY

01101 LOAD C

01110 MULTIPLY

01111 DIVIDE

10000 (NOT BE USED)

10001 INTRCH. AUGMT

10010 LOAD WI FLAG

10011 ——— SPARE

10100 ADDMAGN TO MEM

10101 INTRCH AUGM TO MEM

10110 STR RNDED

10111 ——— SPARE

11000 ——— SPARE

11001 ——— SPARE

11010 ——— SPARE

11011 ——— SPARE

11100 ——— SPARE

11101 ——— SPARE

11110 ——— SPARE

11111 ——— SPARE

x x 0 x x x x x BINARY ARITH

1 DECIMAL ARITH

0 0 x x x x x x SIGNED - AS IS

01 SIGNED - INVERT

10 UNSIGNED - AS IS (+)

11 UNSIGNED INVERT (-)

* CANDIDATE FOR ELIMINATION

CONNECT
INSTRUCTIONS:

| 24-27 | 51 ——— 59 | |
|---|---|---|
| 0000 | 1000xxxx1 | CONNECT |
| | 1001xxxx1 | CONNECT TO MEM |
| | 1010xxxx1 | CONNECT FORTEST |

xxxx → DEFINE EACH OF THE 16 CONNECTIVES

I/O
INSTRUCTIONS:

| 24-27 | 51 ——— 59 | |
|---|---|---|
| 0000 | 1011xxxx1 | |
| | 0000 | READ |
| | 0001 | WRITE |
| | 0010 | CONTROL |
| | 0011 | LOCATE |
| | 0100 | READ EOS |
| | 0101 | WRITE EOS |
| | 0110 | CONTROL EOS |
| | 0111 | LOCATE EOS |
| | 00000 | (NOT TO BE USED) |
| | 00010 | RELEASE |
| | 00100 | COPY CTRL WD |
| | 00110 ——— | SPARE |
| | 01000 ——— | SPARE |
| | 01010 ——— | SPARE |
| | 01100 ——— | SPARE |
| | 01110 ——— | SPARE |

TRANSMIT
INSTRUCTIONS.

| 24-27 | 51 ——— 59 | |
|---|---|---|
| 0000 | 11000---1 | XMIT WORDS DIRECT COUNT |
| | 11001---1 | SWAP WORDS DIRECT COUNT |
| | 1101xxxx1 | XMIT HALF WORDS IMMED. COUNT |
| | 1110xxxx1 | SWAP WORDS IMMED. COUNT |

xxxx → DEFINES COUNT FOR IMMED. OPNS.

NOTE: XMIT ½ WORDS COUNT IS NOW A MAXIMUM OF 16 AS OPPOSED TO FORMER 32.

SPECIAL FULL WD
INSTRUCTIONS : 24-27 ~ 51 ~~~~~~ 59

0000 1111cccc (NOT TO BE USED)

11110abc1 BRANCH ON BIT

| | |
|---|---|
| xx0 | BRANCH IF OFF |
| xx1 | BRANCH IF ON |
| x0x | LEAVE BIT |
| x1x | SET BIT TO ZERO |
| 0xx | LEAVE BIT |
| 1xx | INVERT BIT |

NOTE: BRANCH ON INDICATOR INSTRUCTION CAN BE DEFINED BY ADDRESS (BIT LOCATION) OF THE BRANCH ON BIT INSTRUCTION — OR A SEPARATE OPN. CODE CAN BE PROVIDED BY MAKING BIT 55 A ONE.

STORE INSTRUCTION COUNTER IS AUTOMATICALLY DEFINED BY THE CODE FOR A FULL WORD INSTRUCTION PRECEDING IDENTIFICATION OF A BRANCH INSTRUCTION CODE (HALF WORD)

AVAILABLE OP CODES AS SPARES:

| 24-27 | 51 ~~~~~~ 59 | |
|---|---|---|
| 0000 | 11111xxxx | 16 CODES |